Case Study 1: University Network Security

A university has a campus network connecting computer laboratories, faculty offices, administrative systems, library services, and student Wi-Fi. Recently, the network administrator noticed repeated unauthorized connection attempts against the university's internal servers. Some computers were also communicating with suspicious external IP addresses.

The university decides to implement multiple security controls:

- A **firewall** is placed at the network boundary to control incoming and outgoing traffic.
- An **IDS** monitors network traffic and generates alerts when suspicious patterns are detected.
- An **IPS** is deployed inline to automatically block certain malicious traffic.
- A **VPN** is provided to faculty members who need secure remote access to internal university resources.
- A **proxy server** is used to control and monitor web access from campus computers.

Questions

1. Which security mechanism primarily controls traffic according to predefined rules?
2. Which mechanism detects suspicious activity and generates alerts without necessarily blocking it?
3. Which mechanism can automatically prevent detected malicious traffic?
4. Why would a VPN be useful for faculty working remotely?
5. What role does the proxy server play in this scenario?
6. Identify one limitation of relying only on a firewall.
7. Explain why multiple security mechanisms are used instead of a single security mechanism.

Case Study 2: Online Banking Organization

A bank operates web servers, employee systems, databases, ATMs, and remote offices. Because financial information is highly sensitive, the bank wants to protect communication and monitor suspicious network activity.

The bank deploys a firewall between the public Internet and its internal network. An IDS monitors network traffic for unusual behavior. An IPS is deployed near critical servers to automatically block certain malicious connections. Employees working from branch offices use a VPN to securely connect to the bank's internal network. A proxy server is deployed to regulate employee Internet access and provide an additional intermediary between internal users and external websites.

During one incident, the IDS detects an unusual number of connection attempts against a server. The security team investigates the alert and determines that the traffic may represent an attack. The IPS subsequently blocks similar traffic.

Questions

1. Why is a firewall important for the bank?
2. What is the difference between the IDS and IPS in this situation?
3. Why is VPN encryption important for employees connecting from branch offices?
4. How does the proxy server act as an intermediary?
5. Why might an organization use both IDS and IPS?
6. What could happen if the firewall rules are incorrectly configured?
7. Which security mechanism is most directly associated with secure remote connectivity?

Case Study 3: Manufacturing Company

A manufacturing company operates an industrial network containing production servers, employee computers, monitoring systems, and Internet-connected services. The company wants to reduce unauthorized access and improve monitoring.

The network architecture includes:

- A **firewall** separating the corporate network from the Internet.
- An **IDS** monitoring network traffic for suspicious patterns.
- An **IPS** protecting important network segments from known malicious traffic.
- A **VPN** allowing authorized engineers to remotely access selected company resources.
- A **proxy server** controlling employees' web requests.

The security team discovers that some employees are accessing potentially unsafe websites. At the same time, the IDS reports repeated suspicious traffic directed toward a production server. The organization decides to review its firewall rules, proxy policies, IDS signatures, and VPN access controls.

Questions

1. Which component can regulate employee web access?
2. Which component is primarily responsible for detecting suspicious network activity?
3. Which component can actively block malicious traffic?
4. Which component provides a secure tunnel for remote communication?
5. Why should VPN access be restricted to authorized users?
6. Why should firewall and proxy rules be reviewed periodically?
7. Explain how these four technologies can work together as a layered security architecture.

1. What is the primary purpose of a network firewall?

A. To physically repair damaged network cables
B. To control network traffic according to security rules
C. To increase monitor resolution
D. To compress all network files

2. Which of the following best describes an IDS?

A. A system that detects and alerts about suspicious activity
B. A system used only for encrypting files
C. A system that automatically replaces a router
D. A system used exclusively for remote printing

3. What is a major difference between an IDS and an IPS?

A. IDS detects and alerts, while IPS can detect and actively block traffic
B. IDS encrypts traffic, while IPS decrypts traffic
C. IDS provides VPN connectivity, while IPS provides proxy services
D. There is no difference between them

4. What is the main purpose of a VPN?

A. To provide a secure communication channel over an untrusted network
B. To increase the physical speed of a processor
C. To replace every firewall in an organization
D. To remove the need for authentication

5. Which technology commonly acts as an intermediary between a client and an external web server?

A. Proxy server
B. IDS
C. Keyboard
D. Database server

6. Which device or system can enforce rules such as "allow HTTPS traffic but block unauthorized inbound connections"?

A. Firewall
B. Monitor
C. Printer
D. Scanner

7. An IDS primarily performs which function?

A. Detection and notification
B. Physical destruction of malicious hardware
C. Password generation only
D. Data compression

8. An IPS is generally positioned so that it can:

A. Observe and potentially block network traffic
B. Only display web pages
C. Store employee documents
D. Replace the operating system

9. Which technology is particularly useful when an employee needs secure access to an organization's internal resources from home?

A. VPN
B. Proxy server only
C. Printer
D. Network switch only

10. What does a proxy server primarily do?

A. Acts as an intermediary for requests between clients and external services
B. Physically encrypts every hard disk
C. Replaces the Internet connection
D. Detects every possible malware sample

11. Which of the following is an example of a firewall rule?

A. Allow traffic to a specified service while blocking unauthorized connections
B. Increase CPU clock speed
C. Change screen brightness
D. Format a hard disk

12. A security administrator wants to receive an alert whenever network traffic matches a known suspicious pattern. Which technology is particularly appropriate?

A. IDS
B. VPN
C. Proxy
D. File compression utility

13. Which technology is designed to provide confidentiality for network communication through an encrypted connection?

A. VPN
B. IDS
C. Hub
D. Proxy cache only

14. A company wants to restrict employees from accessing certain categories of websites. Which technology can assist with this requirement?

A. Proxy server
B. IDS alone
C. VPN alone
D. Network cable

15. What does "defense in depth" mean in network security?

A. Using multiple layers of security controls
B. Using only one powerful password
C. Removing all network monitoring
D. Allowing all traffic by default

16. Which security technology can automatically take action against detected malicious traffic?

A. IPS
B. IDS only
C. Monitor
D. Keyboard

17. Why should firewall rules be carefully configured?

A. Incorrect rules can unintentionally allow or block traffic
B. Firewalls automatically eliminate every security vulnerability
C. Firewall rules affect monitor brightness
D. Firewall configuration has no effect on network security

18. Which of the following is a common use of a proxy server?

A. Web traffic control and monitoring
B. Physical network cable repair
C. CPU manufacturing
D. Operating-system installation only

19. A remote employee connects to an organization's internal network through an encrypted tunnel. Which technology is most likely being used?

A. VPN
B. IDS
C. Proxy server
D. Firewall rule only

20. Which statement about an IDS is correct?

A. It can monitor network activity and generate security alerts
B. It always blocks every suspicious packet automatically
C. It is primarily designed to provide encrypted remote access
D. It replaces the need for authentication

21. Which statement about an IPS is correct?

A. It can detect malicious activity and take preventive action
B. It only provides website caching
C. It is used exclusively for encrypting email
D. It cannot inspect network traffic

22. Which technology is commonly deployed at a network boundary to control traffic entering or leaving a network?

A. Firewall
B. Word processor
C. Spreadsheet application
D. Printer

23. A security team wants to investigate suspicious traffic without initially blocking it. Which technology would be particularly useful?

A. IDS
B. IPS
C. VPN
D. Proxy cache

24. Which combination represents a layered network security architecture?

A. Firewall + IDS/IPS + VPN + Proxy
B. Keyboard + Mouse + Monitor + Printer
C. Compiler + Text editor + Calculator + Browser
D. CPU + RAM + Keyboard + Speaker

25. Which statement best summarizes the relationship among firewalls, IDS/IPS, VPNs, and proxy servers?

A. They perform different security functions that can complement one another
B. They all perform exactly the same function
C. Only VPNs are relevant to network security
D. Proxy servers automatically replace all other security mechanisms